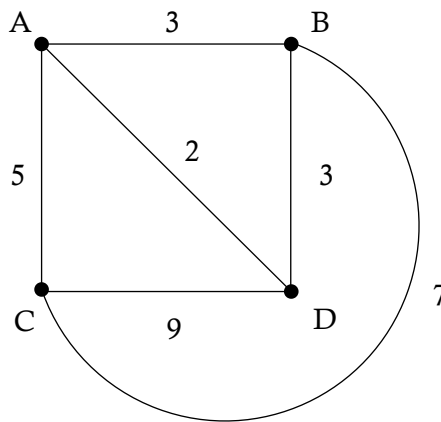


KRUSKAL'S ALGORITHM

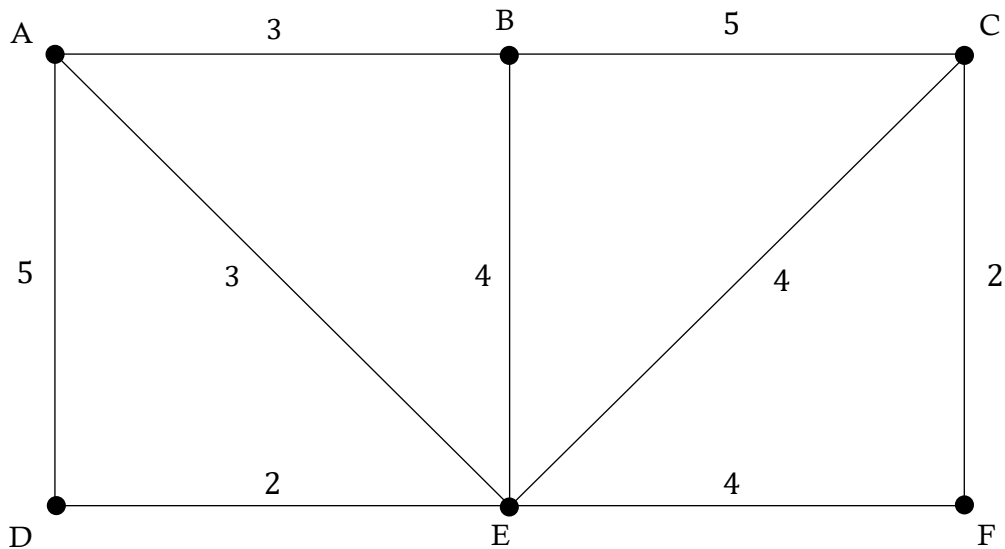
05/12/25

1. Show how Kruskal's algorithm can find both of the minimum spanning trees of the following network:



2. For the network shown below, use Kruskal's algorithm to explain why it is not possible for both CE and EF to be in the same minimum spanning tree, and explain why BE could never be in a minimum spanning tree.

How many different possible minimum spanning trees are there? Draw each one.



3.
 - a. If a network has a unique edge of least weight then explain why that edge must be included in a minimum spanning tree.
 - b. If a network has a unique edge of greatest weight, is it the case that that edge cannot be included in a minimum spanning tree? Provide either a proof or a counterexample.

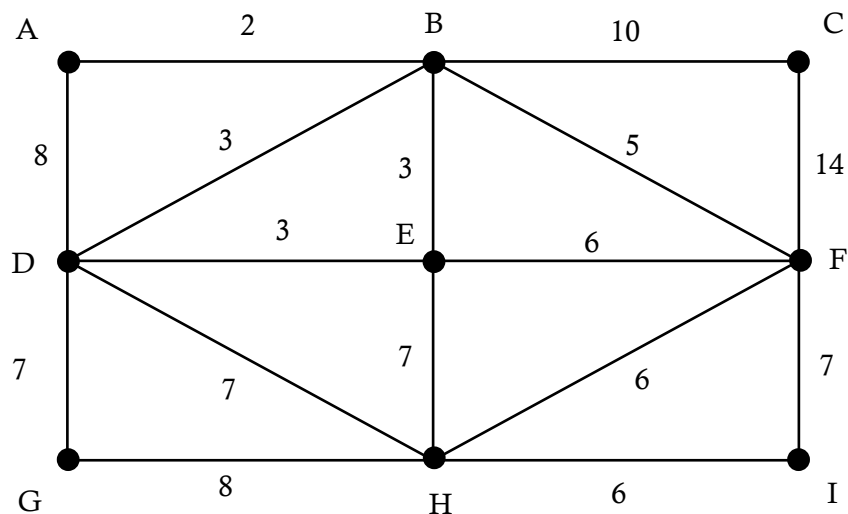
4. Imagine you are programming a computer to apply Kruskal's algorithm. How will you program the computer to check whether the addition of an edge will create a cycle? This is easy for us to do visually in small cases, but a computer cannot 'see' a graph.

One approach is to utilise a particular 'data structure', which takes the form of a table with dynamic columns. Once you have ordered the edges of the graph in order of increasing size, start by putting your first edge in the first column of a table. For the second edge, if it connects to one of the vertices of the first edge, then put it in the first column of the table. Otherwise put it in the second column of the table. For each edge thereafter do the following:

- if none of its vertices appear in any column of the table, then put the edge in a new column;
- if exactly one of its vertices appears in some column of the table, then put the edge into that column;
- if one of its vertices appears in one column of the table, and the other vertex appears in a different column of the table, then combine those two columns, and add the edge to this newly combined column;
- if both of its vertices appears in the same of the table, do not add it.

Apply this procedure to the three networks above, and explain why it works.

5. In a particular complete network on six vertices every edge has a weight which is a positive integer. Given that the combined weight of all of the edges in this network is 100, find the maximum possible weight of the minimum spanning tree. Draw an example to show that your value is obtainable, and explain why no greater value is possible.
6. How many different minimum spanning trees are there for the following network? Draw each one.



7. [Bonus] How many different spanning trees (not necessarily minimum spanning trees) are there in the network shown in Question 3.